INCH-POUND

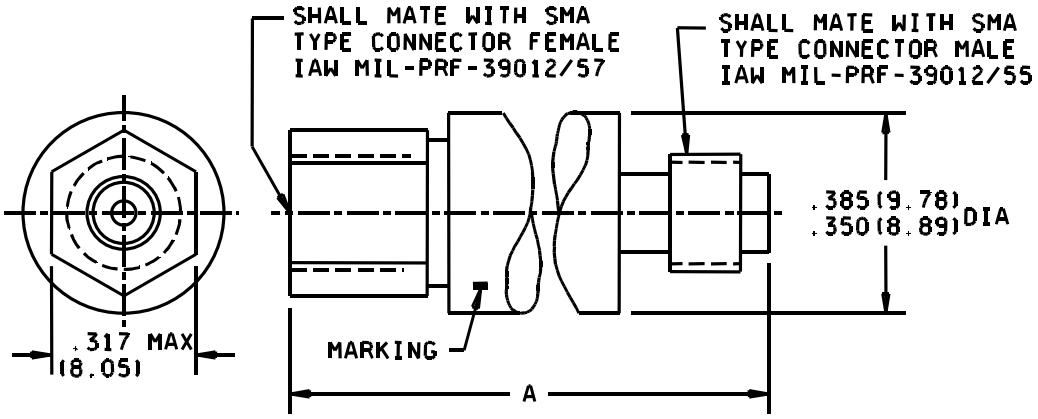
MIL-DTL-3933/14D
w/AMENDMENT 1
8 January 2008
SUPERSEDING
MIL-DTL-3933/14D
7 November 2002

DETAIL SPECIFICATION SHEET

ATTENUATORS, FIXED, COAXIAL LINE (SMA CONNECTORS), FREQUENCY RANGE: DC TO 12.4 GHz, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and the latest issue of MIL-DTL-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

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ENGINEERING DATA:

Attenuators for use in equipment in which precision and stability are secondary considerations, except for temperature sensitivity.

Operating frequency range: dc to 12.4 GHz.

Test frequency range for SWR and attenuation measurements, measured or recorded, when using Automatic Network Analyzer (ANA) or equivalent test equipment:

100 MHz (or lower) to 12.4 GHz.

Basic test frequency for sensitive measurements like connector repeatability, temperature sensitivity, power sensitivity and any other measurements where a 'delta' specification is imposed: 6.0 GHz

Operating temperature range: -55°C to +125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Nominal impedance: 50 ohms.

Attenuation: See table I. Stability and sensitivity:

Maximum change in attenuation after:	up to 10 dB, inclusive (dB)	over 10 dB (dB/dB)
Temperature change or thermal shock	0.5	0.05
Vibration or shock	0.5	0.05
Moisture resistance or salt spray	0.5	0.05
Peak power	0.5	0.05

Frequency sensitivity, maximum: 0.4 dB/dB/GHz.
Temperature sensitivity of attenuation, maximum: 0.0006 dB/dB/°C.

Power: See table I.
Power sensitivity for full input power, maximum: 0.005 dB/dB/Watt.

VSWR: See table I.

Connector repeatability, maximum variation in attenuation: 0.04 dB.

Part or Identifying Number (PIN): M3933/14- (and dash number from table I).

Supersession data: See table II.

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TABLE I. Electrical characteristics.

Dash number		Attenuation dB		Maximum power input at 25°C		VSWR maximum 1/			Weight (oz) max	Dimension A (inches) maximum
		Nominal	Deviation			dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz		
			DC to 12.4 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)					
01	N S	3	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
02	N S	6	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
03	N S	10	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
04	N S	20	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
05	N S	15	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
06	N S	1	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
07	N S	2	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
08	N S	4	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
09	N S	5	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
10	N S	7	±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
11	N S	8	±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
12	N S	9	±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
13	N S	30	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number		Attenuation dB		Maximum power input at 25°C		VSWR maximum 1/			Weight (oz) max	Dimension A (inches) maximum
		Nominal	Deviation			dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz		
			DC to 12.4 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)					
14	N S	40	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)
15	N S	60	±2.0	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)
16	N S	50	±2.0	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)
17	N S	28	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.7	1.850 (46.99)
18	N S	16	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
19	N S	14	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
20	N S	13	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
21	N S	12	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
22	N S	11	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
23	N S	1.5	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
24	N S	31	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.8	1.850 (46.99)

1/ VSWR value is for both ends.

2/ Peak power for a duty cycle of 5×10^{-4}, maximum pulse duration 5 microseconds.

3/ Power input is derated linearly from +25°C to 0.5 watts at +125°C.

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TABLE II. Supersession data.

PIN M3933/14-	Supersedes M3933/15-
01	01
02	02
03	03
04	04
13	05
14	06
13	07
14	08

Referenced documents. In addition to MIL-DTL-3933 this specification sheet references the following documents: MIL-PRF-39012/55 and MIL-PRF-39012/57.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2008-011)

Review activities:
Army - MI
Navy - AS, MC
Air Force - 99

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